

Problem Statement:

Consider the following series: 1,1,2,3,4,9,8,27,16,81,32,243,64,729,128,2187....

This series is a mixture of 2 series : all the odd terms in this series form a geometric series and all the even terms form yet another geometric series.

Write a program to find the Nth term in the series.

In a mysterious library, each book has a unique ID, and some books are quite old. Your task is to find the ID of the oldest book and the average ID of all the books.

- **Input Format**

- A list of integers representing book IDs.

- **Output Format**

- Two integers: the ID of the oldest book and the average ID (rounded down)

- **Example**

- Input: [1001, 1002, 999, 1003]

- Output: (999, 1001)

A supermarket maintains a pricing format for all its products. A value N is printed on each product. When the scanner reads the value N on the item, the product of all the digits in the value N is the price of the item. The task here is to design the software such that given the code of any item N the product (multiplication) of all the digits of value should be computed (price).

- **Example 1:**

- **Input:**

- 5244 -> Value of N

- **Output**

- 160 -> Price

- **Explanation:**

- From the input above

- Product of the digits 5,2,4,4

- $5 \times 2 \times 4 \times 4 = 160$

- Hence, output is 160.